Why pair the four subspaces?

Let $A \in \mathbb{R}^{m \times n}$ and $r = \operatorname{rank}(A)$. Each pair below shares an ambient space.

$$C(A^T), N(A) \subseteq \mathbb{R}^n,$$
$$C(A), N(A^T) \subseteq \mathbb{R}^m.$$

Their dimensions sum to n and m, respectively.

$$\dim C(A^T) + \dim N(A) = r + (n - r) = n,$$
$$\dim C(A) + \dim N(A^T) = r + (m - r) = m.$$

These pairs are orthogonal as well as dimensionally complementary. Every input decomposes uniquely into row-space and nullspace components. Every output decomposes into column-space and left-nullspace components.

Fix the ambient space before taking inner products. A vector in $\mathbb{R}^m$ and one in $\mathbb{R}^n$ cannot be paired by the ordinary dot product unless the dimensions and coordinate interpretation agree.

Why row space is orthogonal to nullspace

If $x \in N(A)$, then $Ax = 0$. Expanding by rows gives

$$\begin{bmatrix} r_1^T \\ \vdots \\ r_m^T \end{bmatrix} x = \begin{bmatrix} r_1^T x \\ \vdots \\ r_m^T x \end{bmatrix} = 0.$$

Thus x is orthogonal to each row. Linearity makes it orthogonal to every linear combination of rows.

$$x^T(c_1 r_1 + \cdots + c_m r_m) = 0.$$

Conversely, orthogonality to every row gives $Ax = 0$. This proves $N(A) = C(A^T)^\perp$. The spaces are orthogonal and their dimensions sum to n, so each is the other's orthogonal complement. Hence

$$C(A^T) = N(A)^\perp.$$

Replacing A by A^T gives $C(A) = N(A^T)^\perp$.

$U^\perp$ consists of all vectors orthogonal to every vector of U. Orthogonality plus the dimension sum establishes the full complement relation.

Orthogonality and complements

Write $U \perp W$ when $u^T w = 0$ for every $u \in U, w \in W$. If $z \in U \cap W$, then $z^T z = 0$, forcing $z = 0$. Therefore $U \cap W = \{0\}$. The converse is false.

In a finite-dimensional inner-product space, if

$$U \cap W = \{0\}, \qquad \dim U + \dim W = \dim V,$$

then $V = U \oplus W$, a direct sum. The subspaces can still meet at an oblique angle. An orthogonal direct sum also requires $U \perp W$.

For example, $U = \operatorname{span}\{(1,0)^T\}$ and $W = \operatorname{span}\{(1,1)^T\}$ intersect only at zero and their dimensions sum to 2. They are not perpendicular. Replacing W by $\operatorname{span}\{(0,1)^T\}$ gives an orthogonal complement.

The inner product determines the orthogonal complement. Complex spaces require conjugate transposes.

Dimensions and a decomposition example

Since $C(A^T) \oplus N(A) = \mathbb{R}^n$, every $x \in \mathbb{R}^n$ has the unique decomposition

$$x = x_{\text{row}} + x_{\text{null}}, \qquad x_{\text{row}}^T x_{\text{null}} = 0.$$

Take

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

whose rank is 2. Its two independent rows form a row-space basis, while the nullspace is spanned by

$$(-2, 1, -1)^T$$

Check directly that

$$(1,2,0)(-2,1,-1)^T = 0, \qquad (0,1,1)(-2,1,-1)^T = 0.$$

Thus $\mathbb{R}^3 = C(A^T) \oplus N(A)$. Since A has full row rank, $N(A^T) = \{0\}$ and $C(A) = \mathbb{R}^2$.

When a subspace is the whole ambient space, its orthogonal complement is the zero subspace. The resulting decomposition with a zero component is still unique.

Projection onto a line

For nonzero $a \in \mathbb{R}^n$, remove from b its component perpendicular to $L = \operatorname{span}\{a\}$. Write the projection as

$$p = \lambda a$$

and require $e = b - p$ to be perpendicular to the whole line.

$$a^T e = a^T(b - \lambda a) = 0.$$

Since $a^T a > 0$,

$$\lambda = \frac{a^T b}{a^T a}, \qquad p = a\frac{a^T b}{a^T a}.$$

The projection matrix is

$$P = \frac{a a^T}{a^T a}, \qquad p = Pb.$$

For $a = (1,1,1)^T$, $b = (2,1,4)^T$, we have $a^T b = 7$, $a^T a = 3$, and

$$p = \frac{7}{3}(1,1,1)^T, \quad e = b - p = \frac{1}{3}(-1,-4,5)^T.$$

Direct substitution gives

$$p + e = (2,1,4)^T = b, \qquad a^T e = \frac{1(-1) + 1(-4) + 1(5)}{3} = 0.$$

Hence $p^T e = 0$, and

Projection onto a line (continued)

$$\|e\|_2 = \frac{\sqrt{(-1)^2 + (-4)^2 + 5^2}}{3} = \frac{\sqrt{42}}{3}.$$

The same values satisfy the normal equation

$$a^T a\, \lambda = a^T b, \qquad 3 \cdot \frac{7}{3} = 7.$$

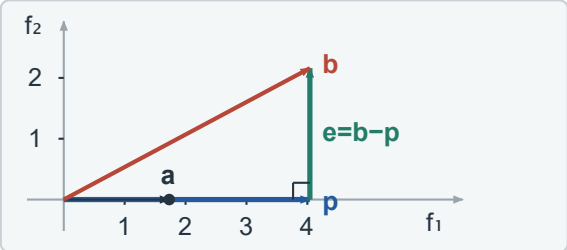
The diagram uses the orthonormal basis $f_1 = a/\sqrt{3}$, $f_2 = e/(\sqrt{42}/3)$ in $\operatorname{span}\{a, e\}$. Its two-dimensional coordinates are

$$[a] = (\sqrt{3}, 0), \quad [p] = (7/\sqrt{3}, 0), \quad [e] = (0, \sqrt{42}/3), \quad [b] = (7/\sqrt{3}, \sqrt{42}/3).$$

These coordinates give an isometry from that plane to $\mathbb{R}^2$, with equal scales on both axes. The right angle represents the original inner product. The short black vector a and longer blue vector $p = (7/3)a$ have distinct endpoints. The green residual is translated to run from p to b.

Line projection: the same numerical example

Exact projection: b=p+e in orthogonal coordinates



Orthonormal axes; equal scale

$f_1=a/\sqrt{3}$ · $f_2=e/(\sqrt{42}/3)$

$a=(\sqrt{3},0)$ · $p=(7/\sqrt{3},0)$

$e=(0,\sqrt{42}/3)$ · $b=p+e$

Original a=(1,1,1), b=(2,1,4)

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Projection onto a subspace

Let $A \in \mathbb{R}^{m\times n}$ have independent columns, and let

$$S = C(A).$$

The projection of b onto S has residual $e = b - Ax$ orthogonal to every basis vector of S.

$$A^T(b - Ax) = 0.$$

This gives the normal equations

$$A^T Ax = A^T b.$$

With full column rank, $A^T A$ is invertible, so

$$p = A(A^T A)^{-1} A^T b.$$

and the projection matrix is

$$P = A(A^T A)^{-1} A^T.$$

The projector P is not A^{-1}, since A need not be square. Nor may $(A^T A)^{-1}$ be replaced by $A^{-1}(A^T)^{-1}$ unless A is square and invertible.

Projection onto a line is the case $n = 1$. Independent columns make $A^T A$ positive definite and invertible because $z^T A^T Az = \|Az\|^2$.

Why column independence is necessary

For every $z \in \mathbb{R}^n$,

$$z^T A^T A z = (Az)^T (Az) = \|Az\|^2 \geq 0.$$

Independent columns imply $Az \neq 0$ for nonzero z. The quadratic form is then strictly positive, so $A^T A$ is positive definite and invertible.

Conversely, dependent columns give a nonzero z with $Az = 0$. Then

$$A^T A z = A^T 0 = 0,$$

so $A^T A$ is singular. In complex spaces replace transpose by conjugate transpose.

Rank deficiency does not destroy uniqueness of the projected vector. It creates redundant parameters, so multiple coefficient vectors can produce the same projection.

Independent columns are necessary and sufficient for invertibility of $A^T A$. Full row rank does not suffice when $m < n$.

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Algebraic properties of a projector

For full column rank A, let

$$P = A(A^T A)^{-1} A^T.$$

The matrix is symmetric and idempotent.

$$P^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P,$$
$$P^2 = A(A^T A)^{-1}(A^T A)(A^T A)^{-1} A^T = P.$$

Idempotence means that projecting twice changes nothing further. Symmetry distinguishes an orthogonal projector from a general oblique projector. The complementary projector

$$I - P$$

projects onto $C(A)^\perp = N(A^T)$, and $P(I - P) = 0$.

Idempotence alone does not imply orthogonality. Symmetry is essential in the real Euclidean setting.

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The nearest point and its error

Let $p = Pb$ and $e = (I - P)b$. Then

$$b = p + e, \qquad p^T e = 0.$$

Pythagoras gives

$$\|b\|^2 = \|p\|^2 + \|e\|^2.$$

For any $q \in C(A)$, the difference $p - q$ lies in $C(A)$, while e is perpendicular to that space. Therefore

$$\|b - q\|^2 = \|e + (p - q)\|^2 = \|e\|^2 + \|p - q\|^2.$$

The right side is minimized uniquely at $q = p$. This proves the nearest-point property without calculus.

For the standard real inner product, the cross term is $e^T(p - q) = 0$. The standard complex inner product uses $e^H(p - q) = 0$, with squared norm $e^H e$. For any other inner product, both orthogonality and the norm must be defined using that inner product.

Least squares

Even when $Ax = b$ is inconsistent, choose

$$x^\star \in \arg\min_x \|b - Ax\|_2^2.$$

The fitted vector $p = Ax^\star$ is the unique projection onto $C(A)$. Full column rank makes the parameter $x^\star$ unique as well. Rank deficiency gives infinitely many minimizers differing by nullspace vectors.

Minimizing the norm or its square gives the same minimizers because squaring is strictly increasing on $[0, \infty)$. The square avoids a square root and gives a linear gradient equation.

The residual norm $\|b - Ax\|_2$ measures the fitting error. If $b \notin C(A)$, even the minimum residual is nonzero, and the fitted vector differs from b.

Least squares minimizes the sum of squared coordinate errors. The optimal residual is perpendicular to the attainable subspace; it need not vanish.

Deriving the normal equations

Define

$$E(x) = (b - Ax)^T (b - Ax).$$

For any displacement h, expand the square.

$$\begin{aligned} E(x+h) - E(x) = & -2h^T A^T (b - Ax) \\ & + h^T A^T A h. \end{aligned}$$

The gradient is therefore

$$\nabla_x E = -2A^T (b - Ax).$$

At an unconstrained minimum $x^\star$, the first-order change vanishes in every direction h, so

$$A^T (b - Ax) = 0 \Longleftrightarrow A^T A x = A^T b.$$

With full column rank, $h^T A^T A h = \|Ah\|^2 > 0$ for every nonzero h. The quadratic remainder is strictly positive, proving the unique global minimum. Both sides of the normal equations have shape $n \times 1$.

For dependent columns, the normal equations may have multiple solutions. If x, z both solve them, let $q = x - z$. Subtracting gives $A^T A q = 0$. Multiplying by q^T,

$$0 = q^T A^T A q = \|Aq\|^2.$$

so $Ax = Az$. The parameter vectors can differ while the projection point remains unique.

Deriving the normal equations (continued)

A^T is not the projection matrix. The condition $A^T e = 0$ tests orthogonality to every column, equivalently $e \in N(A^T)$.

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Three-point fit: compute the intercept and slope

Fit $y = c + mt$ to $(0,1), (1,2), (2,2)$. Keep the three times fixed and vary only c, m.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ m \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The columns are independent. The normal equations therefore have a unique solution:

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} c \\ m \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad \Longleftrightarrow \quad \begin{cases} 3c + 3m = 5, \\ 3c + 5m = 6. \end{cases}$$

Subtract the first equation from the second: $2m = 1$. Substitution gives $c = (5 - 3/2)/3 = 7/6$. The fitted line is $y = 7/6 + t/2$.

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Three residuals must pass two orthogonality checks

Substitute $c = 7/6, m = 1/2$ into every observation, with $e_i = b_i - p_i$:

Time t	Observed	Predicted	Residual	Squared residual
0	1	$7/6$	$-1/6$	$1/36$
1	2	$5/3$	$1/3$	$1/9$
2	2	$13/6$	$-1/6$	$1/36$

$$A^T e = \begin{bmatrix} e_1 + e_2 + e_3 \\ 0e_1 + 1e_2 + 2e_3 \end{bmatrix} = \begin{bmatrix} -1/6 + 1/3 - 1/6 \\ 1/3 - 2/6 \end{bmatrix} = 0, \qquad \|e\|^2 = \frac{1}{6}.$$

A zero residual sum checks only the constant column. The candidate horizontal line $y = 5/3$ has $\tilde{e} = (-2, 1, 1)^T/3$. Its sum is zero, but $(0,1,2)\tilde{e} = 1 \neq 0$ and $\|\tilde{e}\|^2 = 2/3 > 1/6$.

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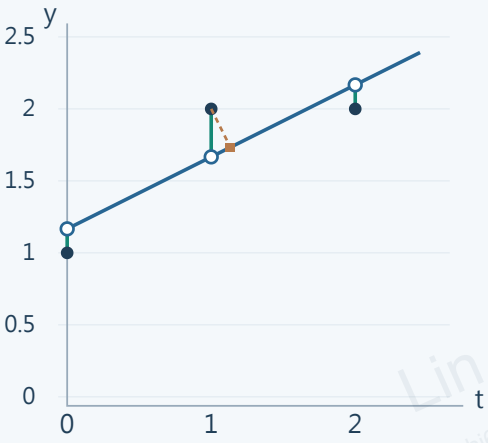
Green segments in the data plot are vertical residuals

The objective is $E(c, m) = \sum_i [b_i - (c + mt_i)]^2$, with every t_i fixed. The middle vertical segment ends at $(1, 5/3)$ and has length $1/3$. The shortest segment ends instead at $(17/15, 26/15)$ and has length $2/(3\sqrt{5})$. These measure different quantities.

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Vertical residuals in the observation plot

Hold t fixed: measure three vertical residuals



Filled dots: observations b_i

Open dots: predictions p_i

Blue line: $y = 7/6 + t/2$

Green: $e_i = b_i - p_i$

$e = (-1/6, 1/3, -1/6)^T$

$\Sigma e_i^2 = 1/6$

Brown dashes: shortest distance

This is a different objective.

Equal units on t and y; the brown segment ends at (17/15, 26/15).

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Orthogonality lives in the three-dimensional data space

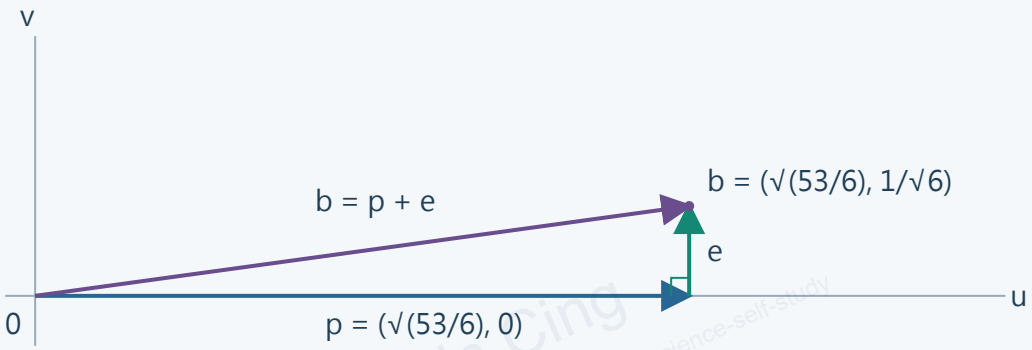
Collect the observations in $b = (1, 2, 2)^T \in \mathbb{R}^3$. Its prediction $p \in C(A)$ and residual $e = b - p$ satisfy $e \perp C(A)$.

In this slice, $\alpha u + \beta v \in C(A)$ exactly when $\beta = 0$. Its horizontal line is only $\text{span}\{p\}$, whereas $C(A)$ is a plane. These data-vector axes differ from the observation plot's time and height axes.

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Orthogonal decomposition in data space

Data-vector space: an isometric slice through p and e



u = p/||p||, v = e/||e||: orthonormal axes, drawn with equal units.
The horizontal line is C(A) ∩ span{p,e} = span{p}, not all of C(A).
$||p||^2 = 53/6$, $||e||^2 = 1/6$, $||b||^2 = 9 = 53/6 + 1/6$.

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Prove a unique minimum, rather than only stationarity

Change the parameters to $c' = 7/6 + \Delta c, m' = 1/2 + \Delta m$. The prediction changes by

$$Ah = (\Delta c, \Delta c + \Delta m, \Delta c + 2\Delta m)^T, \qquad h = (\Delta c, \Delta m)^T.$$

Since $e \perp C(A)$, the cross term vanishes:

$$\begin{aligned} E(c', m') &= \|e - Ah\|^2 = \frac{1}{6} + \|Ah\|^2 \\ &= \frac{1}{6} + 3(\Delta c + \Delta m)^2 + 2(\Delta m)^2. \end{aligned}$$

Both squares vanish only when $\Delta m = 0$ and then $\Delta c = 0$. This proves the unique global minimum. Increasing only the intercept by $1/6$ changes the squared residual norm to $1/6 + 3/36 = 1/4$, which can also be checked observation by observation.

Center time to solve the same example with QR

Set $s = t - 1$. Then $c + mt = a + ms$, where $a = c + m$. The new columns $(1, 1, 1)^T$ and $(-1, 0, 1)^T$ are orthogonal, giving

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ m \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \Rightarrow a = 5/3,\ m = 1/2,\ c = a - m = 7/6.$$

Normalize these columns: $q_1 = (1, 1, 1)^T/\sqrt{3}, q_2 = (-1, 0, 1)^T/\sqrt{2}$. A thin QR factorization of the original matrix is

$$A = QR, \quad Q = [q_1\ q_2], \quad R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}, \quad Q^T b = \begin{bmatrix} 5/\sqrt{3} \\ 1/\sqrt{2} \end{bmatrix}.$$

Solving $Rx = Q^T b$ again gives $m = 1/2, c = 7/6$. Add $q_3 = (1, -2, 1)^T/\sqrt{6}$. The unfit coordinate is $q_3^T b = -1/\sqrt{6}$; its square is exactly $1/6$. Changing the column basis preserves $C(A)$, the projection and the minimum residual. Chapter 11 treats finite-precision stability separately.

rank-deficient least squares

Take

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

The two columns of A are identical, so its rank is 1. Projection onto their line gives

$$p = \frac{a^T b}{a^T a} a = \frac{7}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 14/5 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} x = \begin{bmatrix} 7 \\ 7 \end{bmatrix},$$

and impose only

$$x_1 + x_2 = \frac{7}{5}.$$

Every parameter vector satisfying this relation produces the same $Ax = p$ and minimizes the error. The residual

$$e = \left(-\frac{2}{5}, \frac{1}{5} \right)^T$$

is orthogonal to $(1, 2)^T$. Thus projection uniqueness and parameter uniqueness are separate properties.

rank-deficient least squares (continued)

For deficient rank, construct the projection from independent columns or solve the normal equations for the full solution family. The pseudoinverse selects the minimum-norm member. Regularization generally changes the objective; do not invert singular $A^T A$.

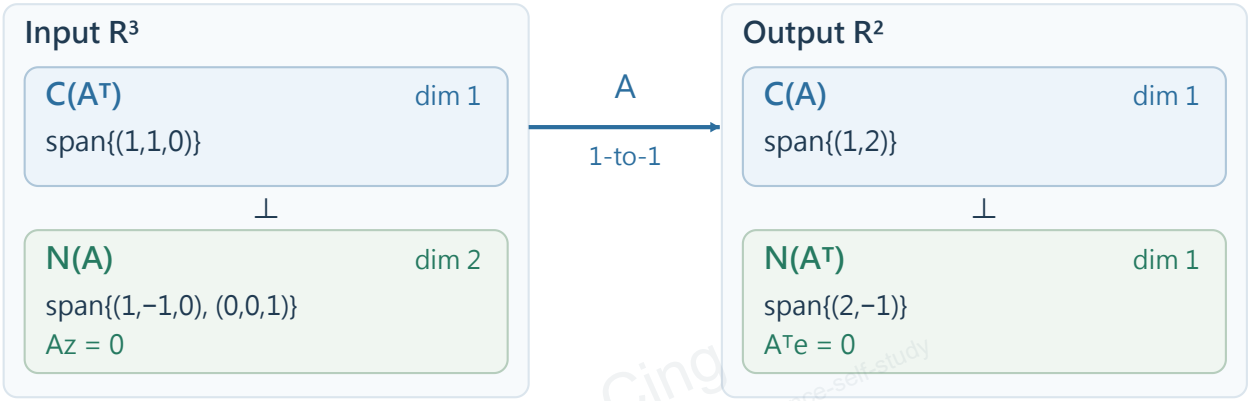
Four subspaces of one rectangular matrix

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix}$, of rank one. Inputs are in $\mathbb{R}^3$; outputs are in $\mathbb{R}^2$.

A maps row space one-to-one onto column space and sends N(A) to zero. Orthogonal residuals lie in $N(A^T)$.

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Four subspaces of a rank-one matrix



Boxes name subspaces and dimensions; their positions are not vector coordinates.

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A unique projection need not give unique parameters

For the same A, take $b = (3, 1)^T$. With $c = (1, 2)^T$, the column space is its span, so

$$p = \frac{c^T b}{c^T c} c = c = (1, 2)^T, \qquad e = b - p = (2, -1)^T.$$

Since $c^T e = 0$ and $p - Ax$ always lies in the column space,

$$\|b - Ax\|^2 = \|e\|^2 + \|p - Ax\|^2 = 5 + \|p - Ax\|^2.$$

The minimum residual norm is $\sqrt{5}$, attained exactly when Ax=p. This requires only $x_1 + x_2 = 1$, with x_3 free. All least-squares solutions are

$$x = \begin{pmatrix} 1/2 + t \\ 1/2 - t \\ s \end{pmatrix}, \qquad t, s \in \mathbb{R}.$$

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One minimum-norm solution among infinitely many solutions

Here $\hat{x} = (1/2, 1/2, 0)^T$ lies in the row space and $z = (t, -t, s)^T$ in the nullspace. Orthogonality gives

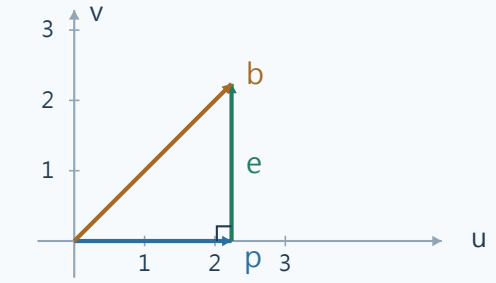
$$x = \hat{x} + z, \qquad \|x\|^2 = \tfrac{1}{2} + 2t^2 + s^2.$$

The plot uses orthonormal axes $u = (1, 2)^T/\sqrt{5}$, $v = (2, -1)^T/\sqrt{5}$. The norm is minimized only at t=s=0. The pseudoinverse in Chapter 7 selects this same $\hat{x}$.

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Projection and minimum-norm solution

Output: orthonormal coordinates



[p]=(√5,0), [e]=(0,√5), [b]=(√5,√5)

All least-squares parameters

x = (1/2+t, 1/2−t, s)

Ax = p = (1,2)

$||x||^2 = 1/2 + 2t^2 + s^2$

t=s=0: x̂ = (1/2,1/2,0)

p is unique; x need not be.

The minimum-norm parameter x̂ is unique.

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Orthonormal bases

Vectors $q_1, \ldots, q_k$ satisfying

$$q_i^T q_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

are orthonormal. Placing them in $Q \in \mathbb{R}^{m \times k}$ gives

$$Q^T Q = I_k.$$

If Q is square, these independent columns form an orthonormal basis of the whole space. Then $Q^{-1} = Q^T$, and Q is an orthogonal matrix.

If Q is rectangular, $Q^T Q = I_k$ gives full column rank, not $QQ^T = I_m$. The two identities have different sizes and different meanings.

Despite the historical name, an orthogonal matrix has orthonormal columns. A rectangular matrix with orthonormal columns has no two-sided inverse.

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Gram–Schmidt construction

For independent vectors $a_1, \dots, a_n$, begin by normalizing

$$q_1 = \frac{a_1}{\|a_1\|}.$$

For a_j, subtract its projections onto the existing directions.

$$z_j = a_j - \sum_{i=1}^{j-1} (q_i^T a_j) q_i,$$

Then normalize

$$q_j = \frac{z_j}{\|z_j\|}.$$

Independence guarantees $z_j \neq 0$. Otherwise a_j would be in the span of its predecessors. For $i < j$, expand $q_i^T z_j$ to obtain

$$q_i^T z_j = q_i^T a_j - \sum_{\ell=1}^{j-1} (q_\ell^T a_j)(q_i^T q_\ell) = q_i^T a_j - q_i^T a_j = 0.$$

Inductively, the first $j-1$ vectors q are orthonormal and span $\operatorname{span}\{a_1, \dots, a_{j-1}\}$. Since a_j equals z_j plus a combination of the earlier directions, replacing z_j by its normalization q_j preserves the expanded span. At every step,

$$\operatorname{span}\{q_1, \dots, q_j\} = \operatorname{span}\{a_1, \dots, a_j\}.$$

Gram–Schmidt construction (continued)

For $a_1 = (1, 1, 0)^T, a_2 = (1, 0, 1)^T$,

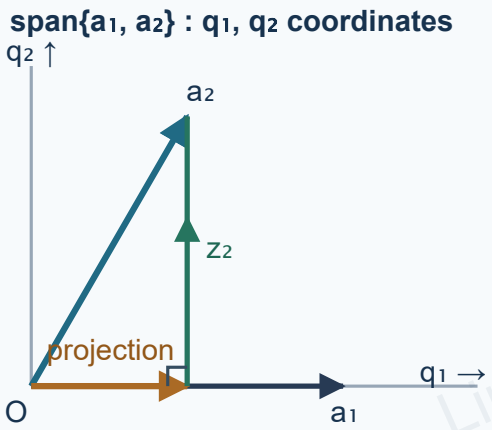
$$q_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T,$$

$$z_2 = a_2 - (q_1^T a_2) q_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T,$$

$$q_2 = \frac{1}{\sqrt{6}}(1, -1, 2)^T.$$

To assess near dependence of directions, compare $\|z_j\|$ to $\|a_j\|$; the absolute length alone confounds direction with scale.

Gram–Schmidt: subtract the projection



$a_2 = (1, 0, 1)^T$

projection = $\frac{1}{2}(1, 1, 0)^T$

$z_2 = \frac{1}{2}(1, -1, 2)^T$

$q_1^T z_2 = 0$

a_2 = projection + z_2

The diagram uses orthonormal coordinates q_1, q_2 in the plane spanned by a_1, a_2; the right angle shows the residual is orthogonal to the first direction.

Measure distance relative to scale

In Gram–Schmidt, z_j is the component of a_j left after projecting onto the previous span. For nonzero a_j, a small ratio $\|z_j\|/\|a_j\|$ indicates that its direction is close to that span relative to its own scale.

For $a_1 = (1, 0)^T$, $a_2 = (0, 10^{-12})^T$, the vectors are exactly orthogonal: although $\|z_2\| = 10^{-12}$, the ratio is 1. By contrast, $a_2 = (1, 10^{-12})^T$ has ratio approximately 10^{-12}, so it is close to the span of a_1.

This is a relative geometric test. In finite precision, input uncertainty and algorithmic stability also matter; a single threshold does not establish exact dependence.

QR and least squares

Collect the Gram–Schmidt expansions into

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular. The coefficients $R_{ij} = q_i^T a_j$ for $i < j$, and $R_{jj} = \|z_j\|$, give $A = QR$ by column multiplication. In the preceding example,

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

For full column rank, least squares reduces to

$$R^T Q^T Q R x = R^T Q^T b \Longrightarrow Rx = Q^T b.$$

Solve the invertible upper-triangular system in R. Reduced Q is $m \times n$. In practice Householder QR avoids forming $A^T A$ and is generally more stable; classical Gram–Schmidt can lose orthogonality when columns are nearly dependent.

Do not cancel Q^T arbitrarily. In the normal-equation derivation one can cancel the invertible factor R^T, since full column rank guarantees invertible R.

A complete QR least-squares example

Use the preceding

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

and $b = (1, 2, 0)^T$. First compute

$$Q^T b = \left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{6}} \right)^T.$$

Solve $Rx = Q^T b$. The second equation gives

$$\sqrt{\frac{3}{2}}\, x_2 = -\frac{1}{\sqrt{6}} \quad \Longrightarrow \quad x_2 = -\frac{1}{3}.$$

The first equation is

$$\sqrt{2}\, x_1 + \frac{1}{\sqrt{2}} \left(-\frac{1}{3} \right) = \frac{3}{\sqrt{2}},$$

so $x_1 = 5/3$. The projection is

$$p = Ax = \left(\frac{4}{3}, \frac{5}{3}, -\frac{1}{3} \right)^T,$$

and the residual is

A complete QR least-squares example (continued)

$$e = b - p = \frac{1}{3}(-1, 1, 1)^T.$$

Check directly that

$$a_1^T e = 0, \qquad a_2^T e = 0,$$

Thus $A^T e = 0$ and $\|e\|^2 = 1/3$. The nonzero residual makes the normal-equation orthogonality condition observable.

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Extension: approximate a function by a line

Prerequisites: definite integration and function-space inner products. Let $V = C[0, 1]$, with real inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)\,dx, \qquad W = \operatorname{span}\{1, x\}.$$

Find $p(x) = c + mx \in W$ minimizing $\int_0^1 (e^x - p(x))^2\,dx$. This is an **integrated squared error**, not pointwise or maximum-error minimization.

Orthogonalize $1, x$ to $u_1 = 1$, $u_2 = x - \frac{1}{2}$. Since $\langle u_1, u_2 \rangle = 0$ and $\langle u_2, u_2 \rangle = \frac{1}{12}$, the two projection coefficients can be computed independently.

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Integrate each term to find the projection

$$\langle e^x, 1\rangle = e - 1, \quad \langle e^x, x - \tfrac{1}{2}\rangle = \tfrac{3-e}{2}.$$

Hence

$$p(x) = (e-1) + 6(3-e)(x - \tfrac{1}{2}) = (4e - 10) + (18 - 6e)x.$$

The residual is orthogonal to u_1, u_2, hence also to their span. For any $q \in W$, Pythagoras gives $\|e^x - q\|^2 = \|e^x - p\|^2 + \|p - q\|^2$, proving uniqueness.

Subtract the squared norms of the two orthogonal components from $\|e^x\|^2 = (e^2 - 1)/2$. The minimum error is

$$20e - \frac{7e^2 + 57}{2} \approx 0.00394022.$$

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Chapter review

- The projection residual is orthogonal to the subspace, yielding the nearest point.
- Normal equations characterize least squares; a unique projection need not mean unique parameters.
- $P = P^T = P^2$; thin Q satisfies $Q^TQ = I$, but generally not $QQ^T = I$.
- Gram–Schmidt and QR preserve column space; numerical stability depends on the algorithm.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 4.

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